

```
mysql> /* data redundancy
/*> having same data at multiple place
/*> data redundancy leads to anomalies [errors]
/*> very hard to maintain database
/*> */
mysql>
mysql> /* types of anomalies
/*> insertion, updation, deletion */
mysql>
mysql> /* Normalization is the process of organizing data in a database to reduce
data redundancy and improve data integrity.
/*> it involves dividing large tables into smaller ones and defining the
relationships between the keys */
```

```
mysql>
mysql>
mysql> CREATE TABLE student_details (
->     student_id INT,
->     student_name VARCHAR(50),
->     courses VARCHAR(100),
->     trainers VARCHAR(100),
->     trainer_phones VARCHAR(100)
-> );
```

Query OK, 0 rows affected (0.09 sec)

```
mysql> INSERT INTO student_details
-> VALUES
-> (101, 'Ravi', 'Java, Python', 'Kumar, Ramesh', '9876543210, 9876543211'),
-> (102, 'Priya', 'Python, SQL', 'Ramesh, Suresh', '9876543211, 9876543212'),
-> (103, 'Arun', 'Java, SQL', 'Kumar, Suresh', '9876543210, 9876543212');
```

Query OK, 3 rows affected (0.02 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from student_details;
```

```
+-----+-----+-----+-----+
--+
| student_id | student_name | courses      | trainers      | trainer_phones
|
+-----+-----+-----+-----+
--+
|      101 | Ravi         | Java, Python | Kumar, Ramesh | 9876543210,
9876543211 |
|      102 | Priya        | Python, SQL  | Ramesh, Suresh | 9876543211,
9876543212 |
|      103 | Arun         | Java, SQL    | Kumar, Suresh  | 9876543210,
9876543212 |
+-----+-----+-----+-----+
```

```
--+
3 rows in set (0.00 sec)
```

```
mysql> create table student_course_1nf(
```

```

-> student_id int,
-> student_name varchar(30),
-> course varchar(30),
-> trainer varchar(30),
-> trainer_phone varchar(15)
-> );

```

Query OK, 0 rows affected (0.11 sec)

```

mysql> insert into student_course_1nf values
-> (101,'ravi','python','ramesh','9876543211'),
-> (101,'ravi','java','kumar','9876543210'),
-> (102,'priya','python','ramesh','9876543211'),
-> (102,'priya','sql','suresh','9876543212'),
-> (103,'arun','java','kumar','9876543210');

```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

```

mysql> insert into student_course_1nf values
-> (103,'arun','sql','suresh','9876543212');

```

Query OK, 1 row affected (0.01 sec)

```

mysql> select * from student_course_1nf;

```

student_id	student_name	course	trainer	trainer_phone
101	ravi	python	ramesh	9876543211
101	ravi	java	kumar	9876543210
102	priya	python	ramesh	9876543211
102	priya	sql	suresh	9876543212
103	arun	java	kumar	9876543210
103	arun	sql	suresh	9876543212

6 rows in set (0.00 sec)

```

mysql> create table student_2nf(
-> student_id int primary key,
-> student_name varchar(50)
-> );

```

Query OK, 0 rows affected (0.08 sec)

```

mysql> insert into student_2nf values
-> (101,'ravi'),
-> (102,'priya'),
-> (103,'arun');

```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```

mysql> create table enrollement_2nf(
-> student_id int,
-> course varchar(30),

```

```

-> trainer varchar(50),
-> trainer_phone varchar(15),
-> primary key(student_id,course),
-> foreign key(student_id) references student_2nf(student_id)
-> );

```

Query OK, 0 rows affected (0.03 sec)

```

mysql> insert into enrollement_2nf values
-> (101,'java','kumar','9876543210'),
-> (101,'python','ramesh','9876543211'),
-> (102,'sql','suresh','9876543212'),
-> (102,'python','ramesh','9876543211'),
-> (103,'java','kumar','9876543210'),
-> (103,'sql','suresh','9876543212');

```

Query OK, 6 rows affected (0.01 sec)

Records: 6 Duplicates: 0 Warnings: 0

```
mysql> select * from enrollement_2nf;
```

student_id	course	trainer	trainer_phone
101	java	kumar	9876543210
101	python	ramesh	9876543211
102	python	ramesh	9876543211
102	sql	suresh	9876543212
103	java	kumar	9876543210
103	sql	suresh	9876543212

6 rows in set (0.00 sec)

```
mysql> select * from student_2nf;
```

student_id	student_name
101	ravi
102	priya
103	arun

3 rows in set (0.00 sec)

```

mysql> create table trainer_3nf(
-> trainer_id int primary key,
-> trainer_name varchar(50),
-> trainer_phone varchar(15)
-> );

```

Query OK, 0 rows affected (0.03 sec)

```
mysql> insert into trainer_3nf values
```

```

-> (1,'kumar','9876543210'),
-> (2,'ramesh','9876543211'),

```

```
-> (3,'suresh','9876543212');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> insert into course_3nf values
-> (1,'java',1),
-> (2,'python',2),
-> (3,'sql',3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> select * from trainer_3nf;
+-----+-----+-----+
| trainer_id | trainer_name | trainer_phone |
+-----+-----+-----+
|          1 | kumar        | 9876543210    |
|          2 | ramesh       | 9876543211    |
|          3 | suresh       | 9876543212    |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from course_3nf;
+-----+-----+-----+
| course_id | course_name | trainer_id |
+-----+-----+-----+
|          1 | java        |          1 |
|          2 | python      |          2 |
|          3 | sql         |          3 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> create table enrollement_3nf(
-> student_id int,
-> course_id int,
-> primary key(student_id,course_id),
-> foreign key (student_id) references student_2nf(student_id),
-> foreign key (course_id) references course_3nf(course_id)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> desc enrollement_3nf;
+-----+-----+-----+-----+-----+-----+
| Field      | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| student_id | int  | NO   | PRI | NULL    |       |
| course_id  | int  | NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

```
mysql> insert into enrollement_3nf
```

```

-> values
-> (101,1),
-> (101,2),
-> (102,2),
-> (102,3),
-> (103,1),
-> (103,3);

```

Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0

```
mysql> select * from student_2nf;
```

student_id	student_name
101	ravi
102	priya
103	arun

3 rows in set (0.00 sec)

```
mysql> select * from course_3nf;
```

course_id	course_name	trainer_id
1	java	1
2	python	2
3	sql	3

3 rows in set (0.00 sec)

```
mysql> select * from enrollement_3nf;
```

student_id	course_id
101	1
103	1
101	2
102	2
102	3
103	3

6 rows in set (0.00 sec)

mysql> notee

ER Models

Entity - Relationship Models

components:

-> entity

-> strong entity = rectangle

-> weak entity = double rectangle

-> attributes = ellipse

-> simple attribute

-> composite attribute

-> key attribute = ellipse and underscore

-> multivalued attribute = double ellipse

-> derived attribute = dotted ellipse

-> relations

-> one-to-one | - |

-> one-to-many | - <

-> many-to-one > - |

-> many-to-many > - <